



Black hole action spotted

NASA's space telescope spotted a rare cosmic occurrence - a star being devoured by a black hole. <https://digit.in/oct19-53>

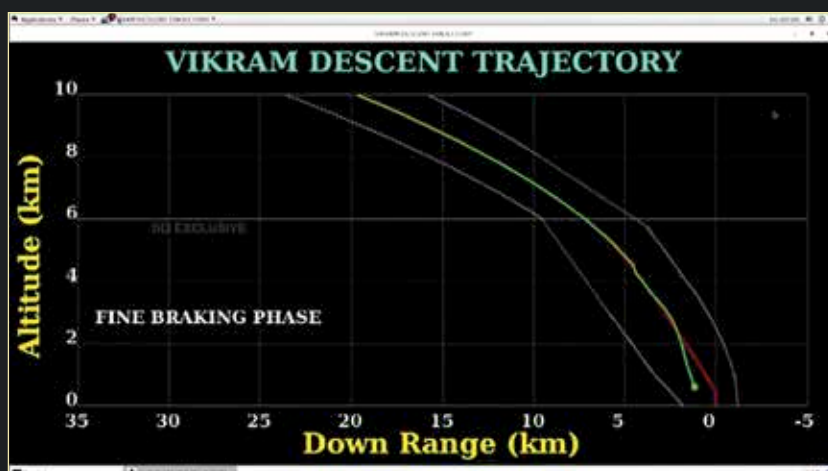


NASA Black Hole simulation

Jeremy Schnittman at NASA used a custom software to create a stunning Black Hole simulation. <https://digit.in/oct19-54>

execute the soft landing were entirely developed by ISRO. The final trajectories for the lander were loaded four hours before the descent operation commenced. There were four phases to the descent, the rough braking phase, the attitude and absolute navigation control phase, the fine braking phase, and the terminal descent phase. The first phase brought down the lander from an altitude of 30 kilometers to 7.4 kilometers, the second phase brought down Vikram to 5 kilometers, the third to 400 meters, and the final phase was meant to gradually execute the touchdown, where the lander only moves vertically. The lander was to only use two of its five thrusters during the terminal descent stage. Just before touchdown, the main engine was meant to fire.

800 N of thrust were generated by the four engines on board during the initial stages of controlled descent. The steerable engines performed admirably during the controlled descent. Unfortunately, in the third phase of the landing, communication was lost with the Vikram lander. The data will have to be analysed to check if the algorithm was responsible for



The actual trajectory (in red) vs the planned one (in green)

best and have always made India proud. These are moments to be courageous, and courageous we will be!" NASA from its official account, tweeted, "Space is hard. We commend @ISRO's attempt to land their #Chandrayaan2 mission on the Moon's South Pole. You have inspired us with your journey and look forward to future opportunities to explore our solar system together."

ISRO was unable to establish contact with the rover before the Lunar sunset. Figures displayed on the landing screen show that just before the transition to the fine braking stage, the lander made a somersault. It also began to deviate from its planned trajectory. When the lander should have been moving into the terminal descent stage, it was moving at 48.1 metres per

second horizontally, and 59 meters per second vertically. The lander was also about a kilometer away from its intended landing spot. At this point, the horizontal velocity should have been zero, and the vertical velocity about 2 meters per second.

On the consoles, the visualisation of the lander was frozen at this juncture. Communication link was lost. ISRO tried to initiate communications

with the lander for the rest of the Lunar day, but was unable to do so. The orbiter of the Chandrayaan 2 mission managed to locate the lander on the surface, but was unable to contact it as well. The problem could have been because of the propulsion system, the control system, or a combination of the two.

One source of possible information is that although the Earth communications were offline, the screen showing the status of the imager data download was still showing "on" till the very end. For most of the time that the lander was communicating back, all the systems on board performed as expected.

The orbiter is functioning as expected, and the science objectives of the mission will continue. ISRO said in a statement, "The Orbiter has already been placed in its intended orbit around the Moon and shall enrich our understanding of the moon's evolution and mapping of the minerals and water molecules in the Polar Regions, using its eight state-of-the-art scientific instruments. The Orbiter camera is the highest resolution camera (0.3m) in any lunar mission so far and shall provide high resolution images which will be immensely useful to the global scientific community. The precise launch and mission management has ensured a long life of almost 7 years instead of the planned one year." 



The lander wobbling

the lander veering off course, or there was some other reason. The four extendable legs on the lander were intended to dampen the impact and protect the sensitive instruments on board, as well as the Pragyaan rover.

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The Prime Minister encouraged the effort, and tweeted, "India is proud of our scientists! They've given their